A multiple changepoint procedure for complex model structures

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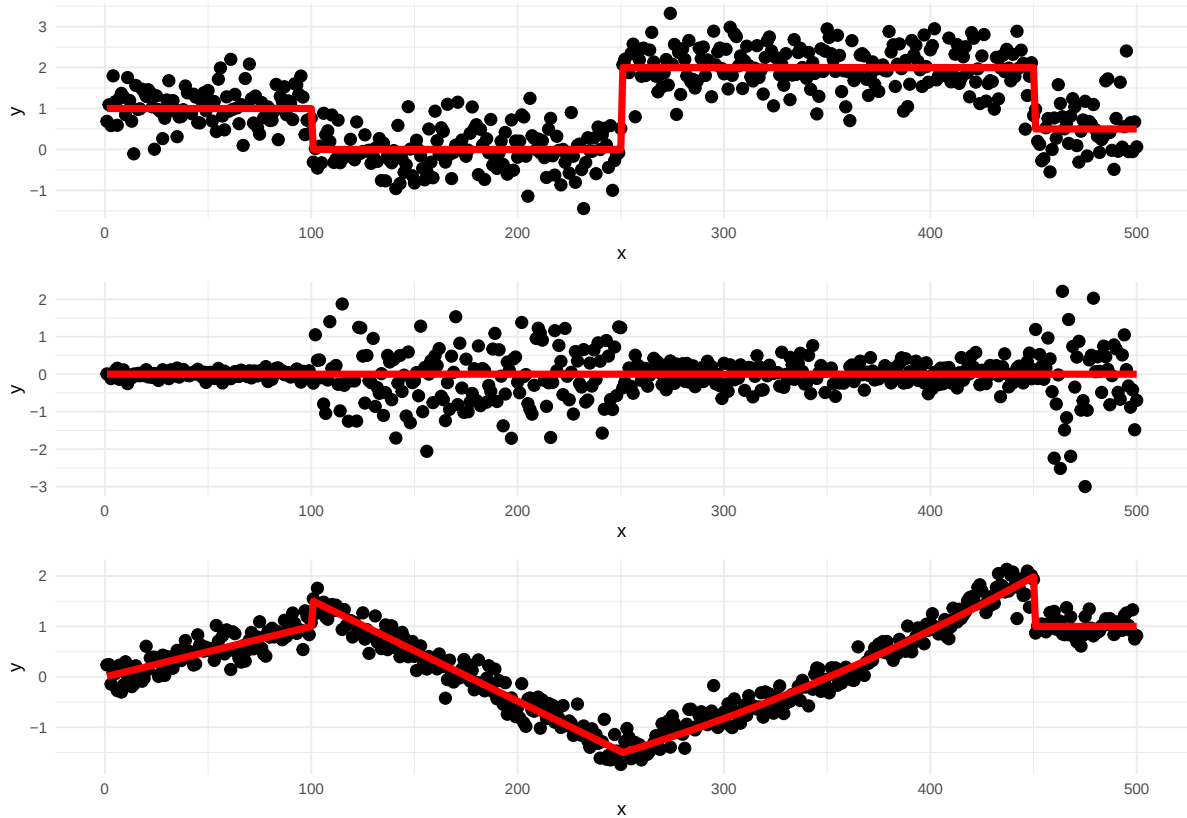
Outline

- What are changepoints?
- What do I mean by complex model structures?
- GD-cpt
- Application to climate data

What are changepoints?

For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.

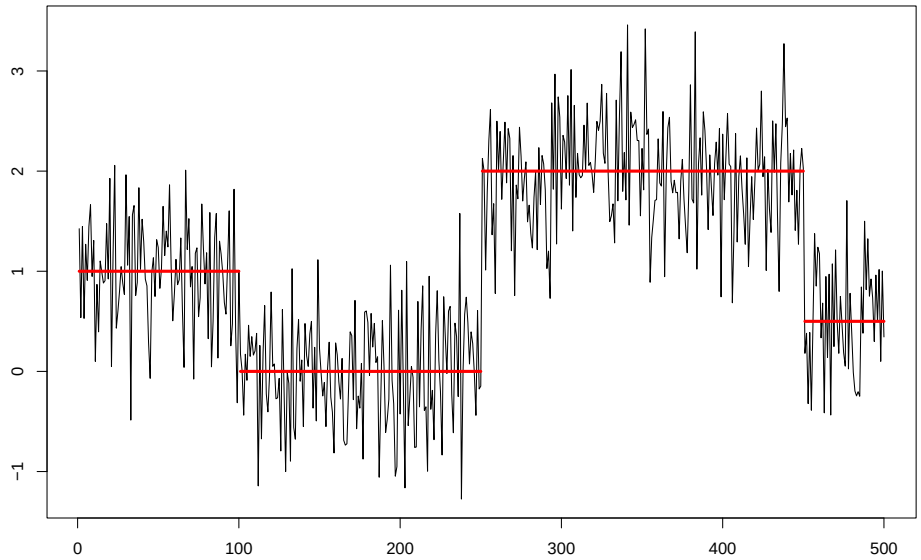


Change in mean

Assume we have time-series data where

$$Y_t|\theta_t \sim \mathrm{N}(\theta_t, 1),$$

but where the means, θ_t, are piecewise constant through time.



Inferring Changepoints

We want to infer the number and position of the points at which the mean changes. One approach:

Likelihood Ratio Test

To detect a single changepoint we can use the likelihood ratio test statistic:

$$LR = \max_{\tau}\{\ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n})\}.$$

We infer a changepoint if $LR > \beta$ for some (suitably chosen) β. If we infer a changepoint its position is estimated as

$$\tau = \arg\max\{\ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n})\}.$$

Extension to multiple changes

Define m to be the number of changepoints, with positions $\boldsymbol{\tau} = (\tau_0, \tau_1, \ldots, \tau_{m+1})$ where $\tau_0 = 0$ and $\tau_{m+1} = n$.

- Multiple change hypothesis test?
- Binary segmentation
- Multiscale windowing

All the above have advantages and disadvantages but are inherently approximations as they conditionally detect changepoints.

Likelihood Ratio Tests

Recall the likelihood ratio test decision for a change:

$$\max_{\tau}\{\ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n})\} > \beta.$$

Then one application of the Likelihood ratio test can be viewed as

$$\min_{m\in\{0,1\},\boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i})\right] + \beta m \right\}$$

Repeated application is thus aiming to minimise

$$\min_{m,\boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[-\ell(y_{\tau_{i-1}:\tau_i})\right] + \beta m \right\}$$

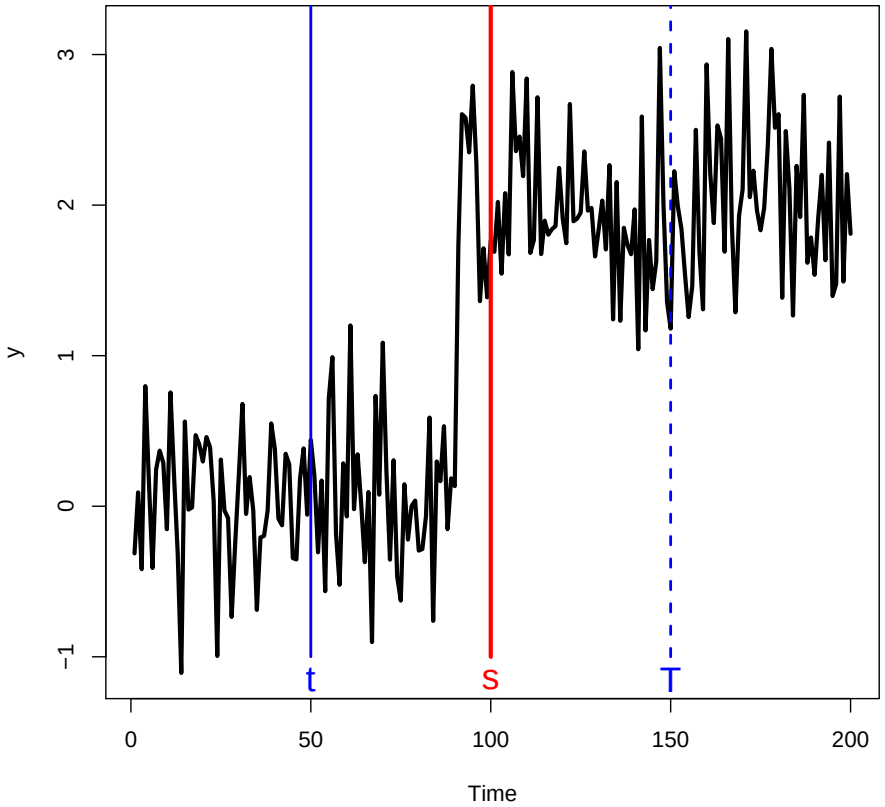
Problem



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

PELT in a nutshell

- Dynamic programming allows us to only worry about the location of the *last* change.
- Pruning means that as we go through the data we are smart about which locations are potential last change locations.

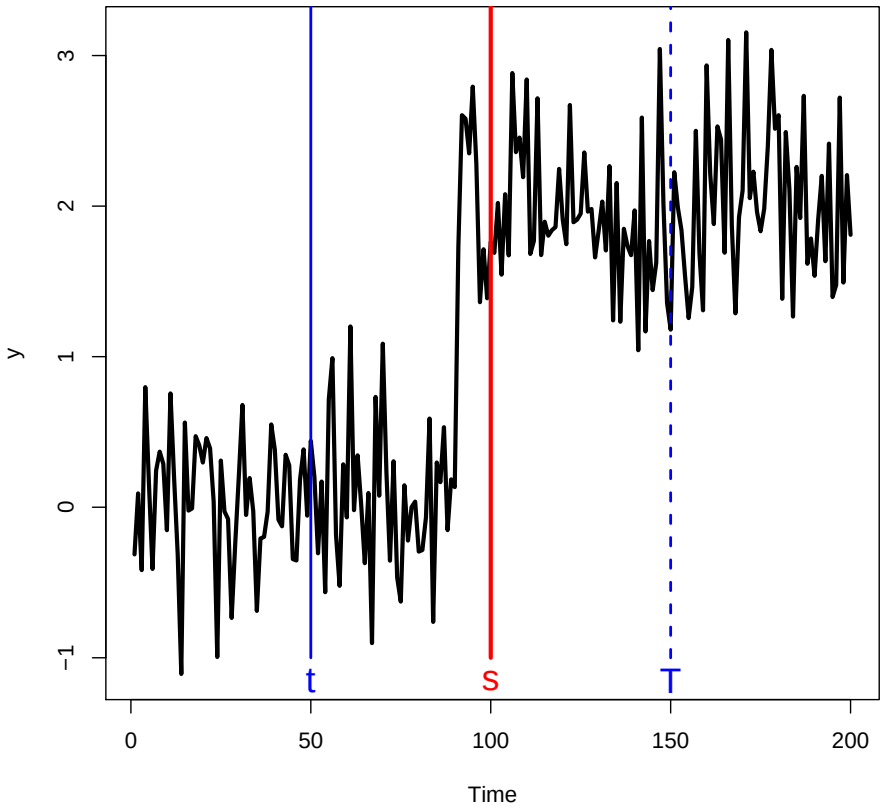


PELT: Pruning

Let $0 < t < s < T$, if

$$F(t) + \mathcal{C}(y_{(t+1):s}) < F(s)$$

then at any future time $T > s$, t can never be the optimal last changepoint prior to T.
We can prove that, under certain regularity conditions, the expected computational complexity will be $\mathcal{O}(n)$.

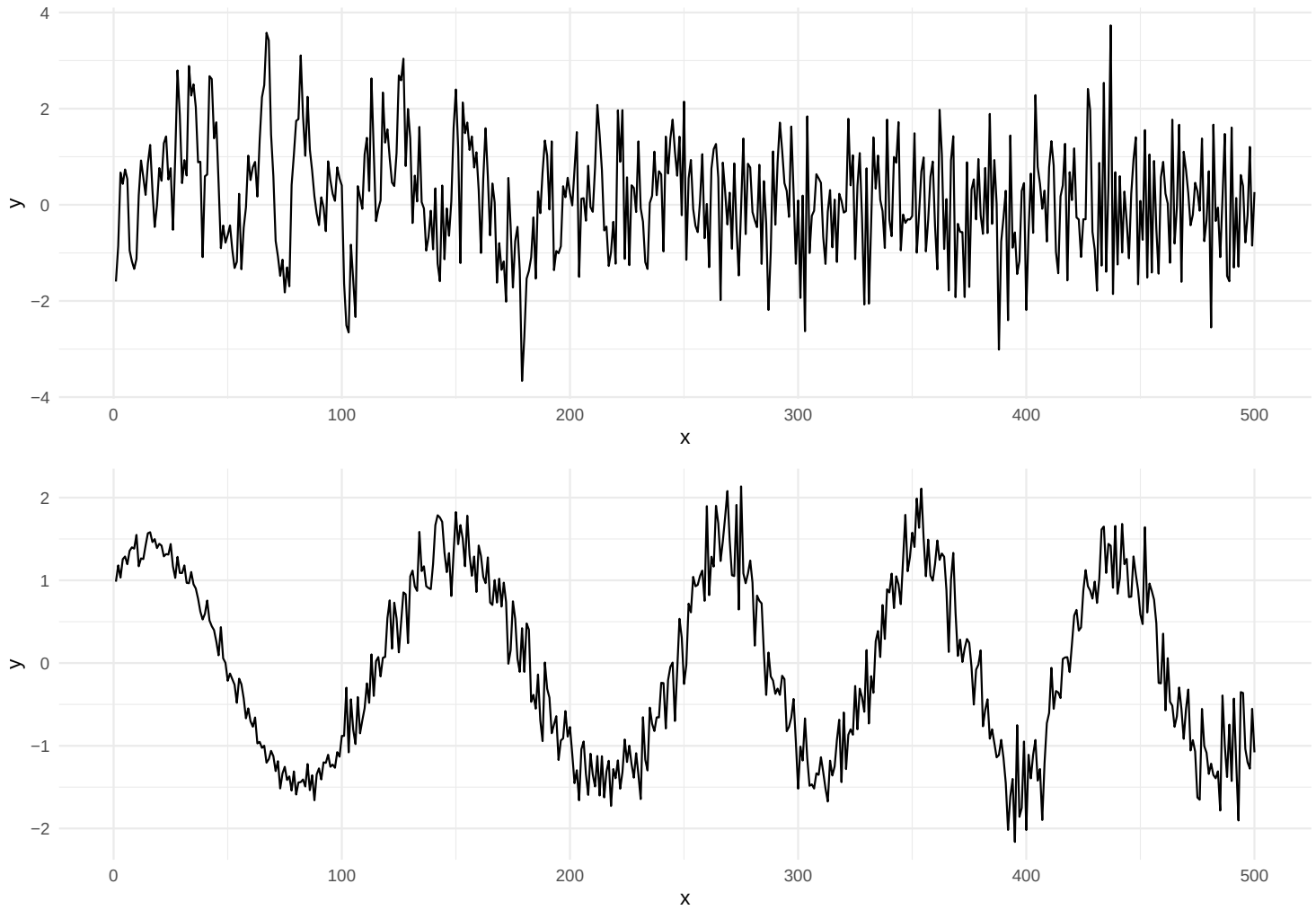


Other options

- PDPA: looks at the optimizer as a function of the (single) parameter
- FPOP: combines PELT and PDPA for single parameter settings
- use computational geometry (convex hull) to compute the solution even faster
- factor in constraints (up-down-up, physics constrained)

All of these approaches solve the optimization problem exactly.

More complicated changes



Interesting Challenges

- Optimization
- Model development
- Theoretical properties
- Limits of detection
- Small sample and large sample in the same data

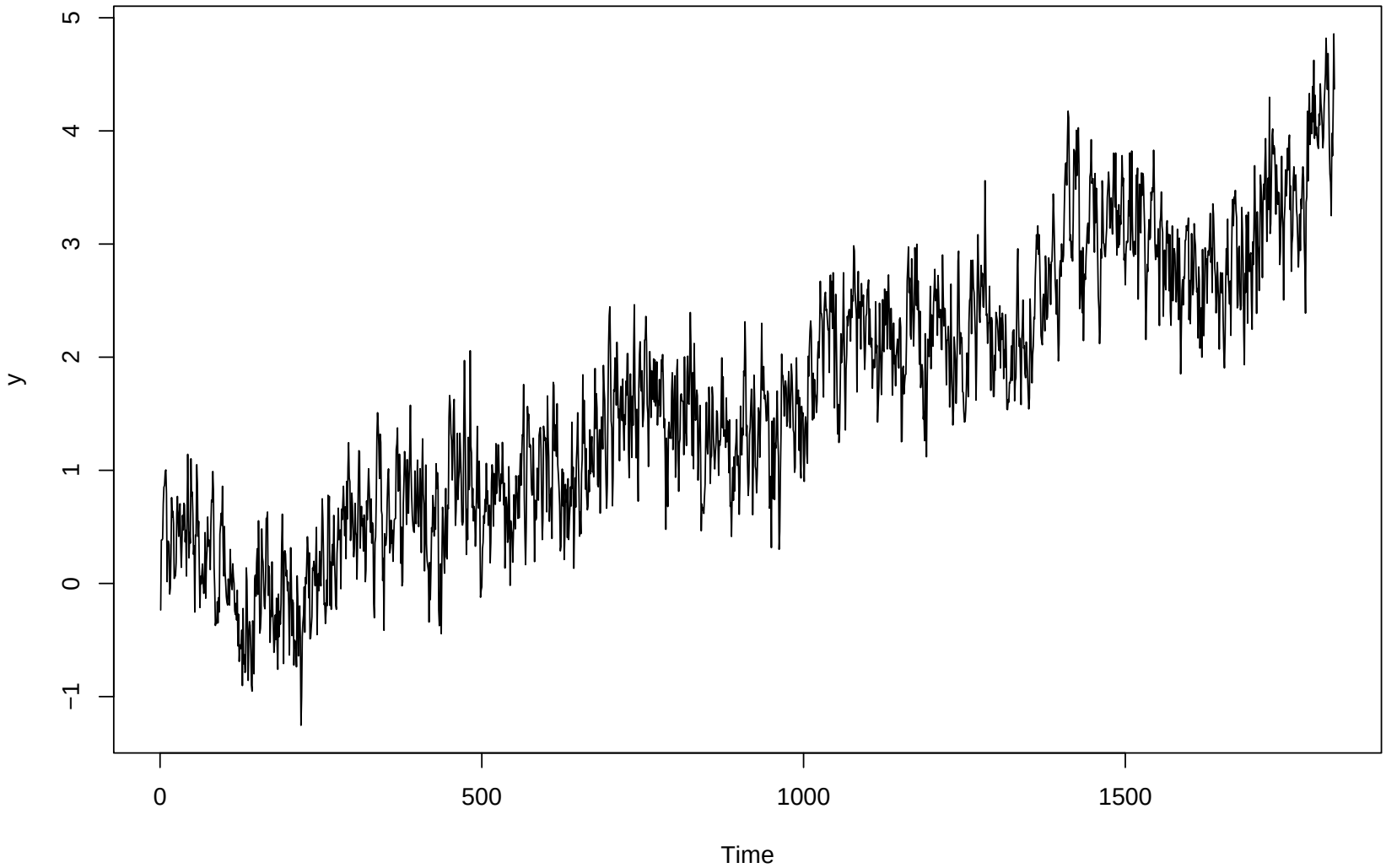


Key practical challenges

- Model building
- What can change?
- What fitting algorithm to use?
- Penalty selection

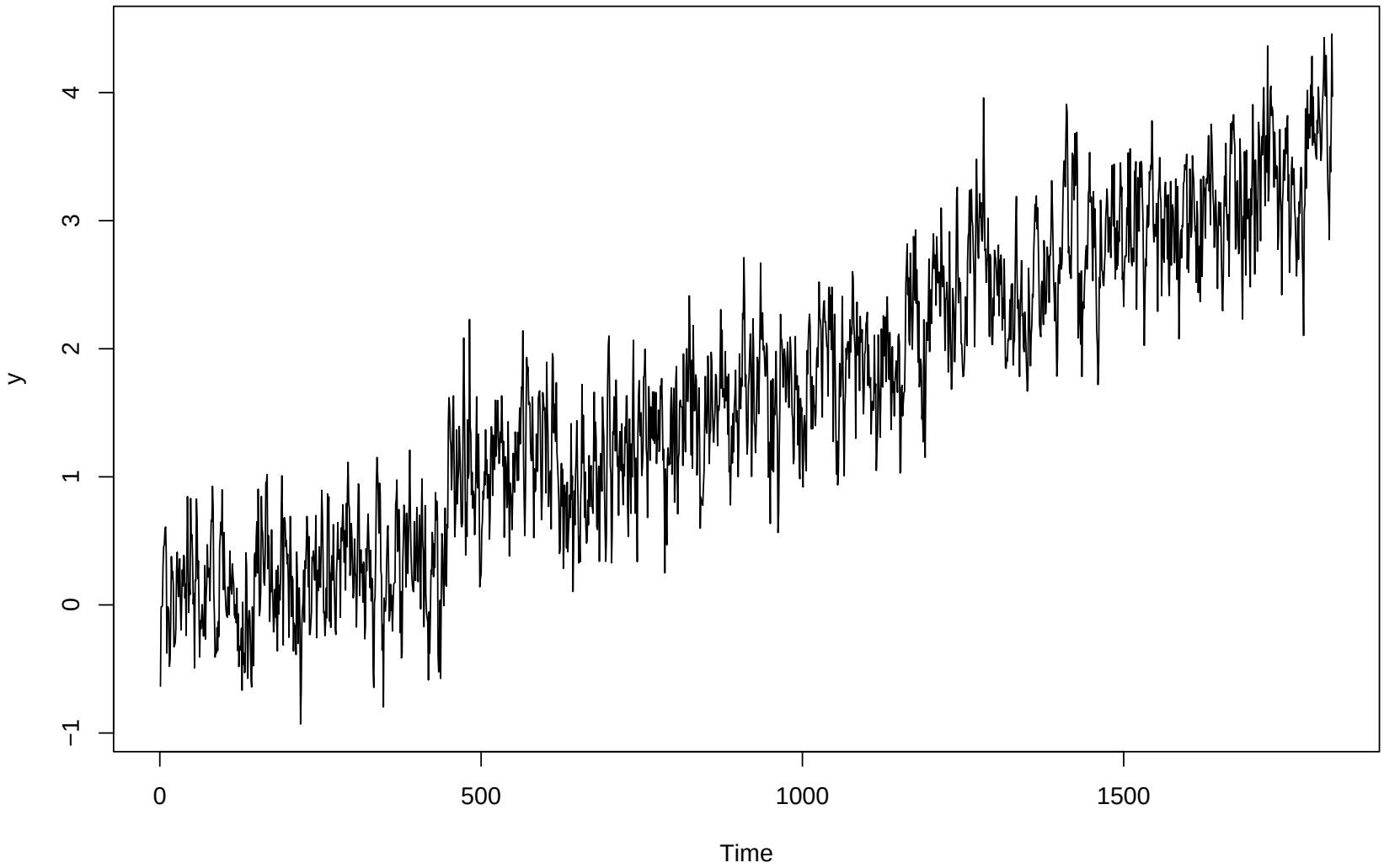
$$\min_{m,\boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[-\mathcal{C}(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

Example data

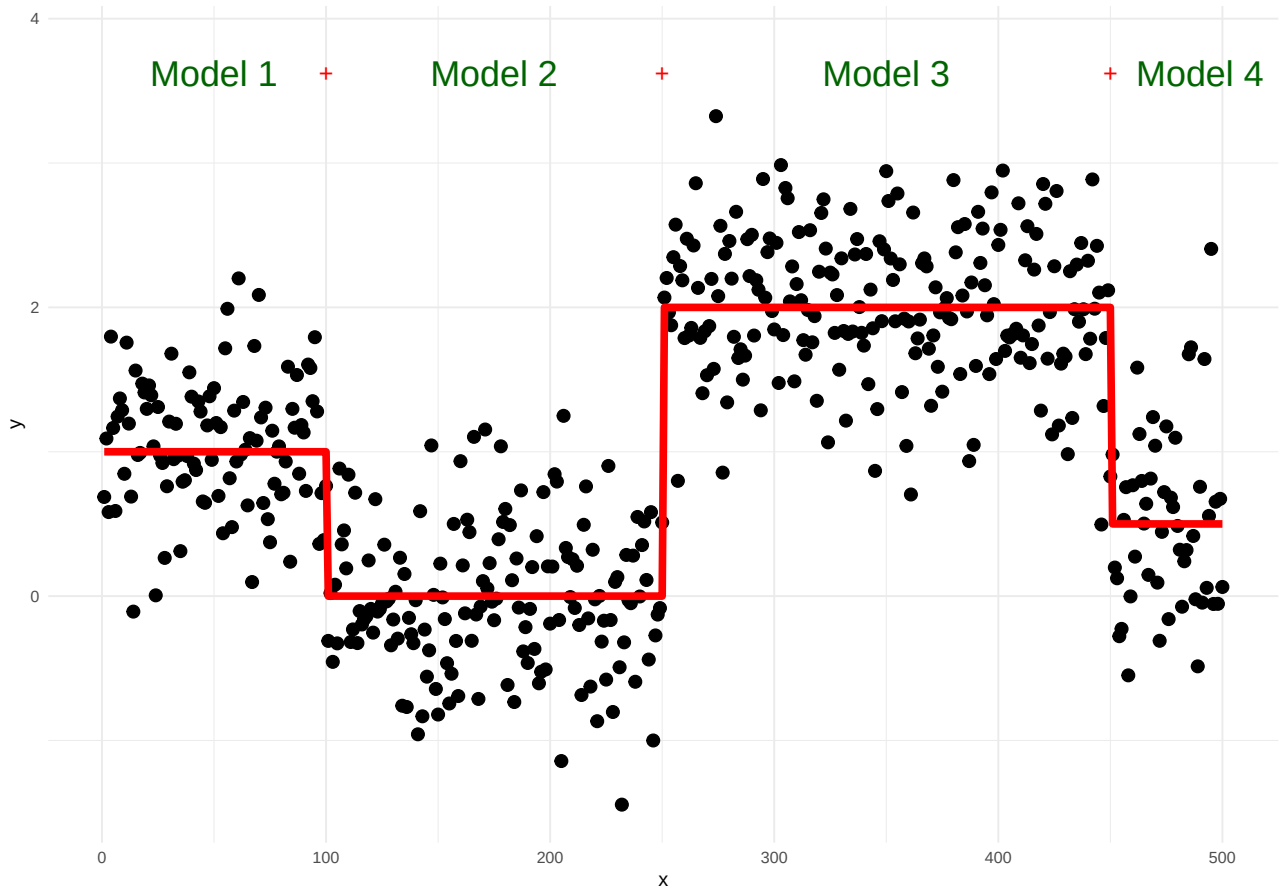


If you wanted to identify changepoints how would you model this?

Example data



Additivity



Global parameters

$$\min_{m,\boldsymbol{\tau}} \left\{ \sum_{i=1}^{m+1} \left[\mathcal{C}(y_{\tau_{i-1}:\tau_i}) \right] + \beta m \right\}$$

How can we identify changes when we have global parameters?

- Allow them to change and fit additive model
  - Not parsimonious
  - Requires post-analysis inference to decide if a change did occur
  - Not suitable for some applications, e.g sub-seasonal changepoints
- Estimate them and remove
  - Robust estimation needed
  - Sensitivity of changepoints to estimates

Ideally we want to fit jointly.

Idea

- If we have a good estimate of global parameters, we know how to fit the changepoints
- If we know the changepoints, we can get a good estimate of global parameters

Given initial estimates, we can iterate between these and converge to a solution.

Problem: potential loops

Solution

1. Initialise global parameters
2. Fit changepoint model to residuals from global parameters
3. Conditional on the changepoints, update global parameters via a gradient step
4. Refit the changepoint model to the new residuals
5. Iterate 3 and 4 until convergence

Does it work?

Let θ_c be changing, and θ_g be global parameters respectively.

Theorem (Finite sample convergence to a fixed point)

For $\{Y_t\}, t \in 1, \ldots, N$, and a changepoint method that guarantees,

$$(\hat{\theta}_c, \hat{\tau}_{1:m}, \hat{m}) = \arg \min_{\tau_{1:m}, m, \theta_c} \left[C(y_{1:N}, \tau_{1:m}, m, \theta_c | \theta_g) + \beta m \right]$$

Then,

1. *GD-cpt converges to a fixed point that depends on the initial $\theta_g^{(0)}$.*
2. *For the global minimizer, $\tilde{\lambda} = (\tilde{\tau}_{1:\tilde{m}}, \tilde{m}, \tilde{\theta}_c, \tilde{\theta}_g)$, if $\theta_g^{(s)} = \tilde{\theta}_g$ then when the changepoint step is completed, $\theta_c^{(s)} = \tilde{\theta}_c$ and the algorithm stops at this global minimum.*

Does it work?

Theorem (Consistency under infill asymptotics)

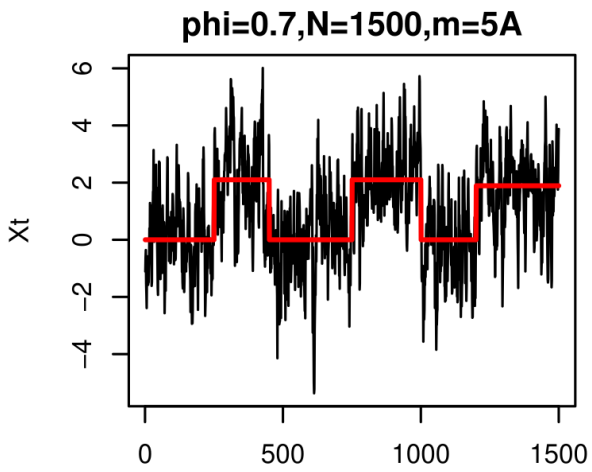
As $N \to \infty$, when there is asymptotic separation between the changepoint times,

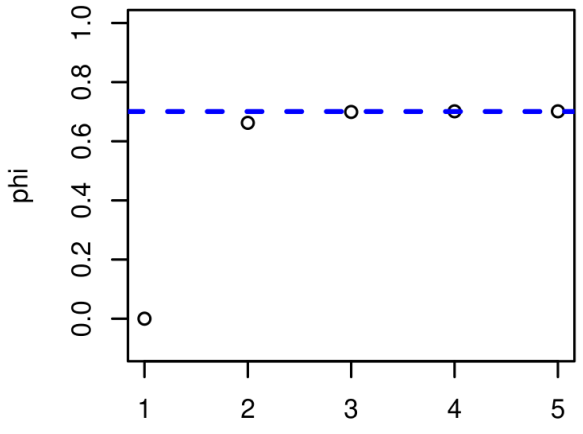
$$\hat{m} \xrightarrow{\mathcal{P}} m, \quad \hat{\tau}_i \xrightarrow{\mathcal{P}} \tau_i \ \text{ for } \ 1 \leq i \leq m.$$

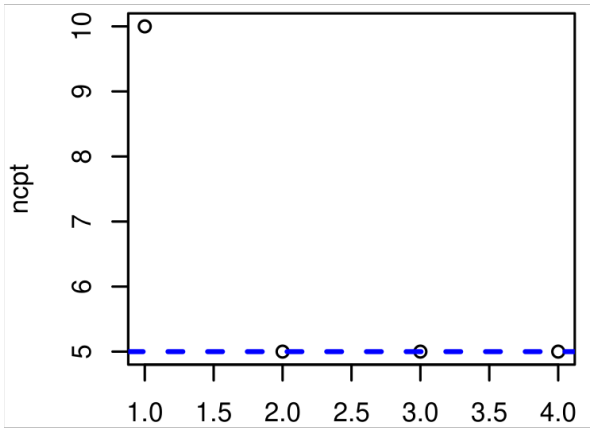
and the global and local parameters are consistently estimated in that

$$(\hat{\beta}_g)_i \xrightarrow{\mathcal{P}} (\beta_g)_i, 1 \leq i \leq d_g; \quad (\hat{\beta}_s)_i \xrightarrow{\mathcal{P}} (\beta_s)_i, 1 \leq i \leq d_s.$$

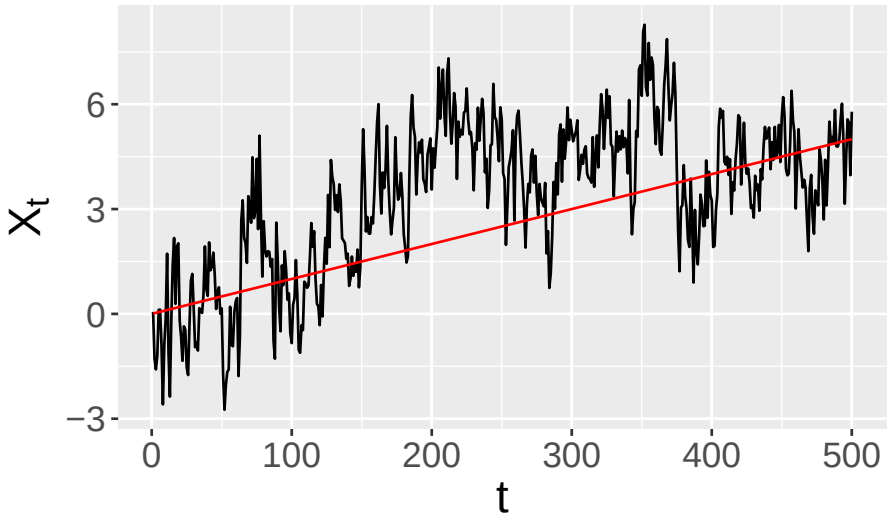
Does it work?

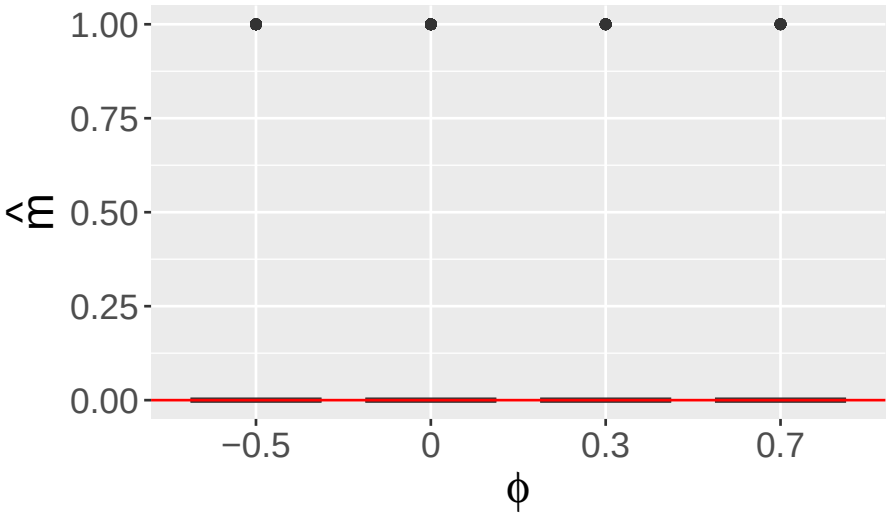

phi=0.7,N=1500,m=5A
Xt
6
4
2
0
−2
−4
0
500
1000
1500

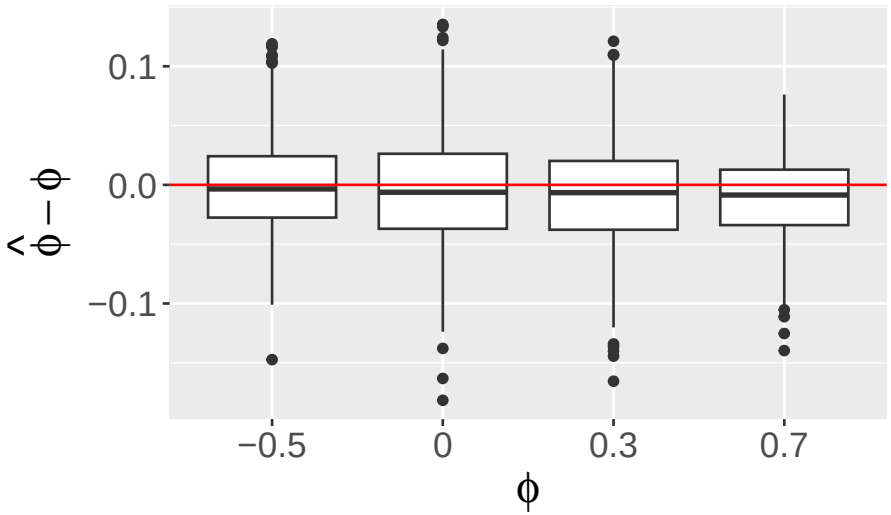

phi
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5

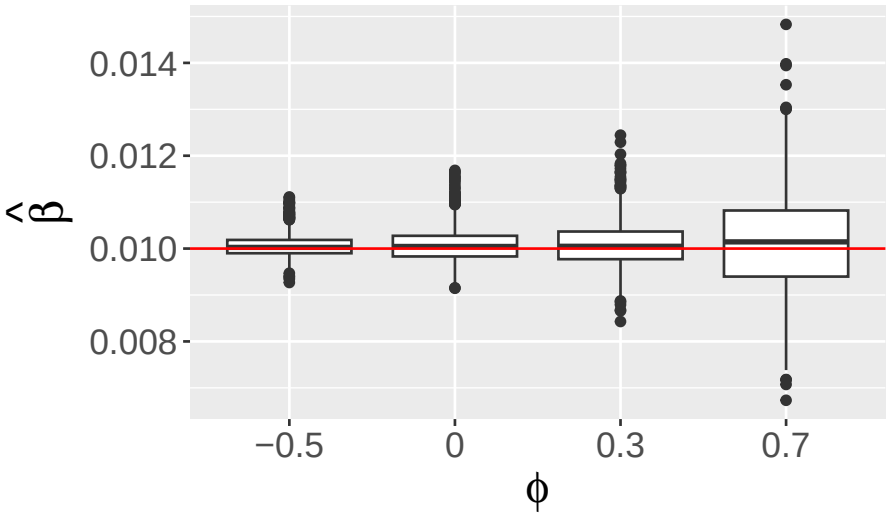

ncpt
10
9
8
7
6
5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

Trend AR1

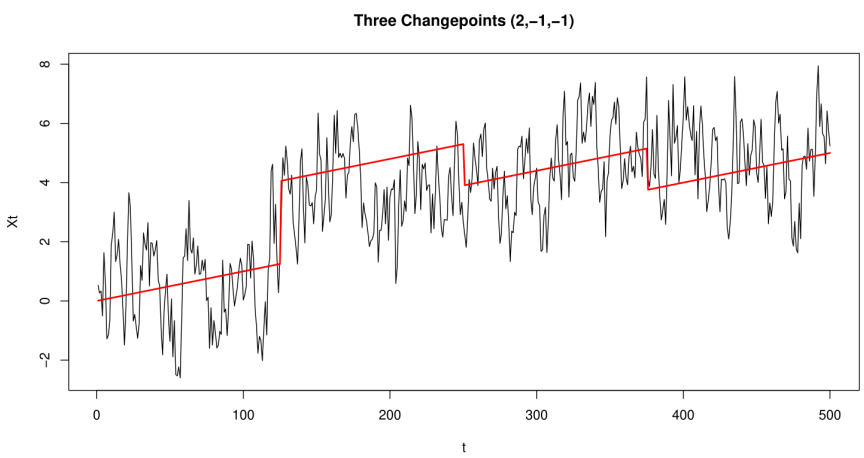

X_t
6
3
0
−3
0
100
200
300
400
500
t


$\hat{m}$
1.00
0.75
0.50
0.25
0.00
−0.5
0
0.3
0.7
ϕ

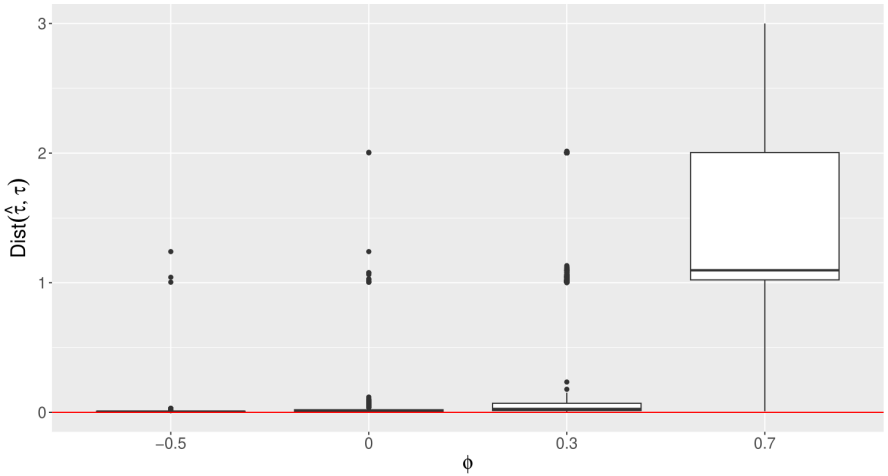

$\hat{\phi} - \phi$
0.1
0.0
−0.1
−0.5
0
0.3
0.7
ϕ


$\hat{\beta}$
0.014
0.012
0.010
0.008
−0.5
0
0.3
0.7
ϕ

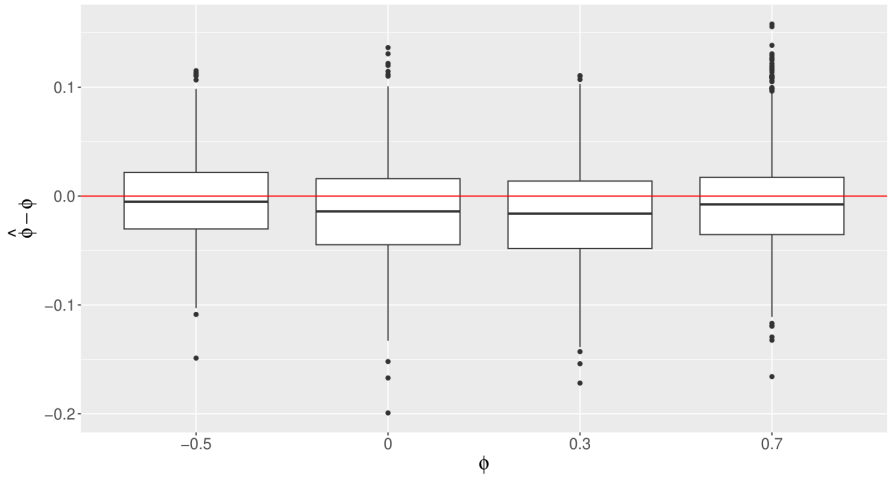
Trend AR1



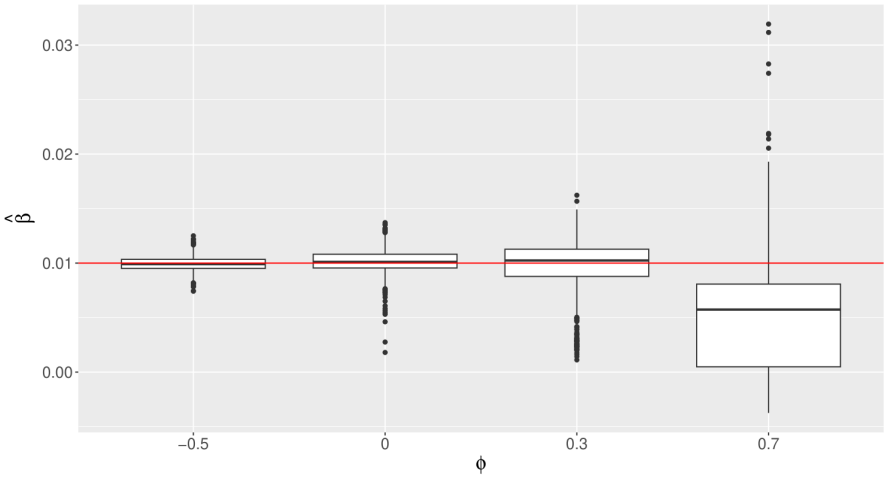
(a) Signal



(b) distance between τ and $\hat{\tau}$

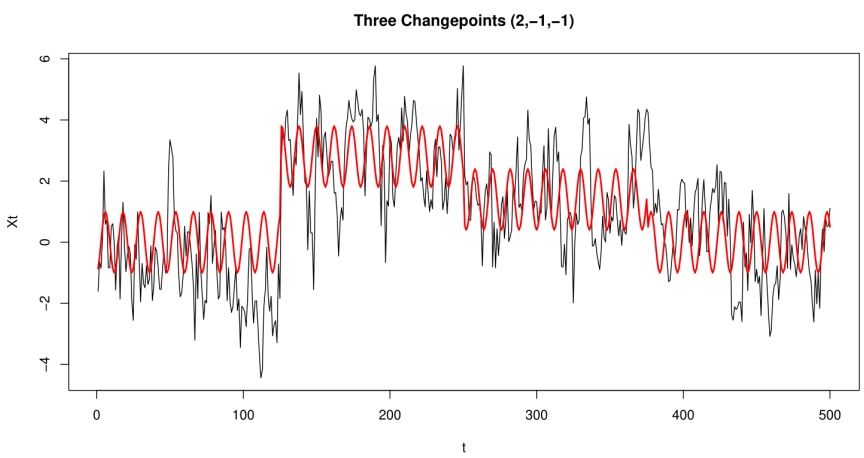


(c) $\hat{\phi} - \phi$

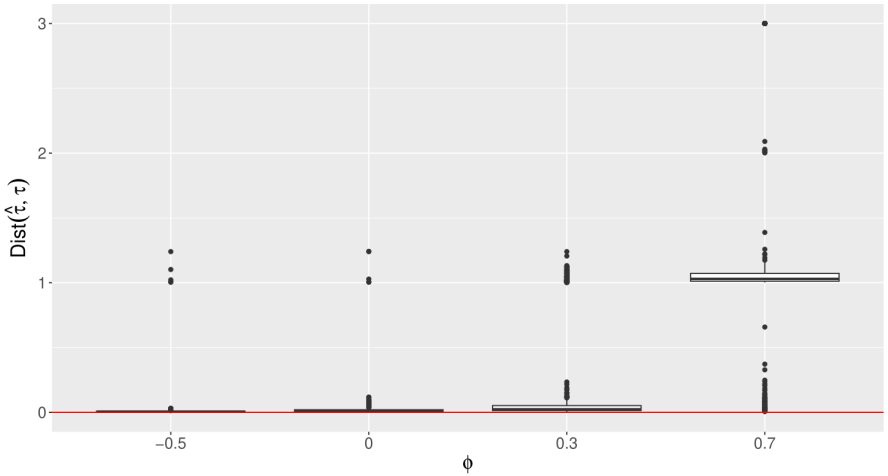


(d) $\hat{\beta}$

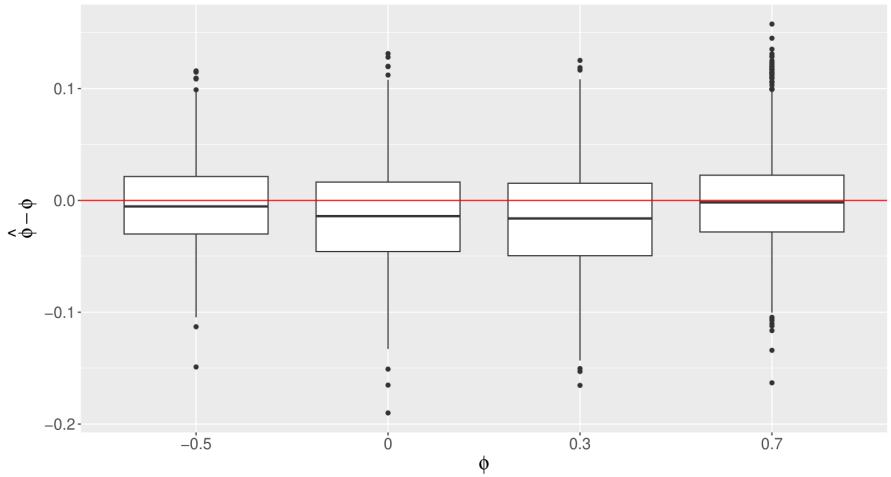
Seasonal AR1



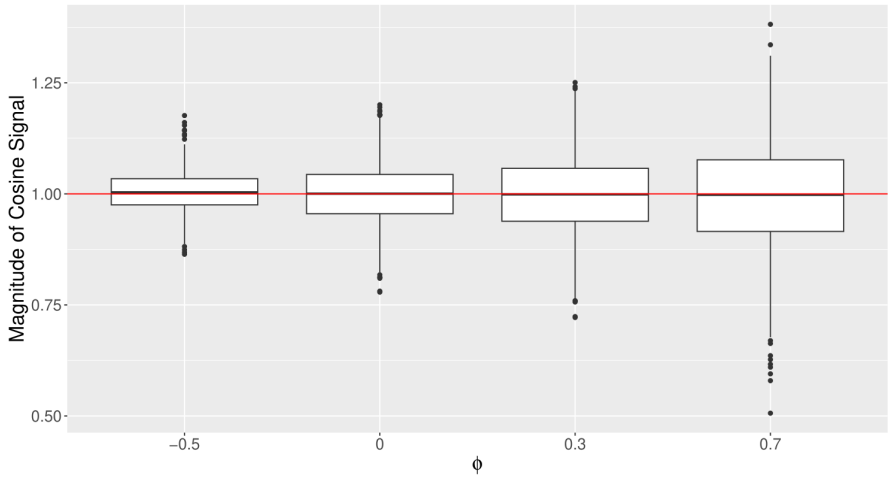
(a) Seasonal signal $T = 12$



(b) distance between τ and $\hat{\tau}$

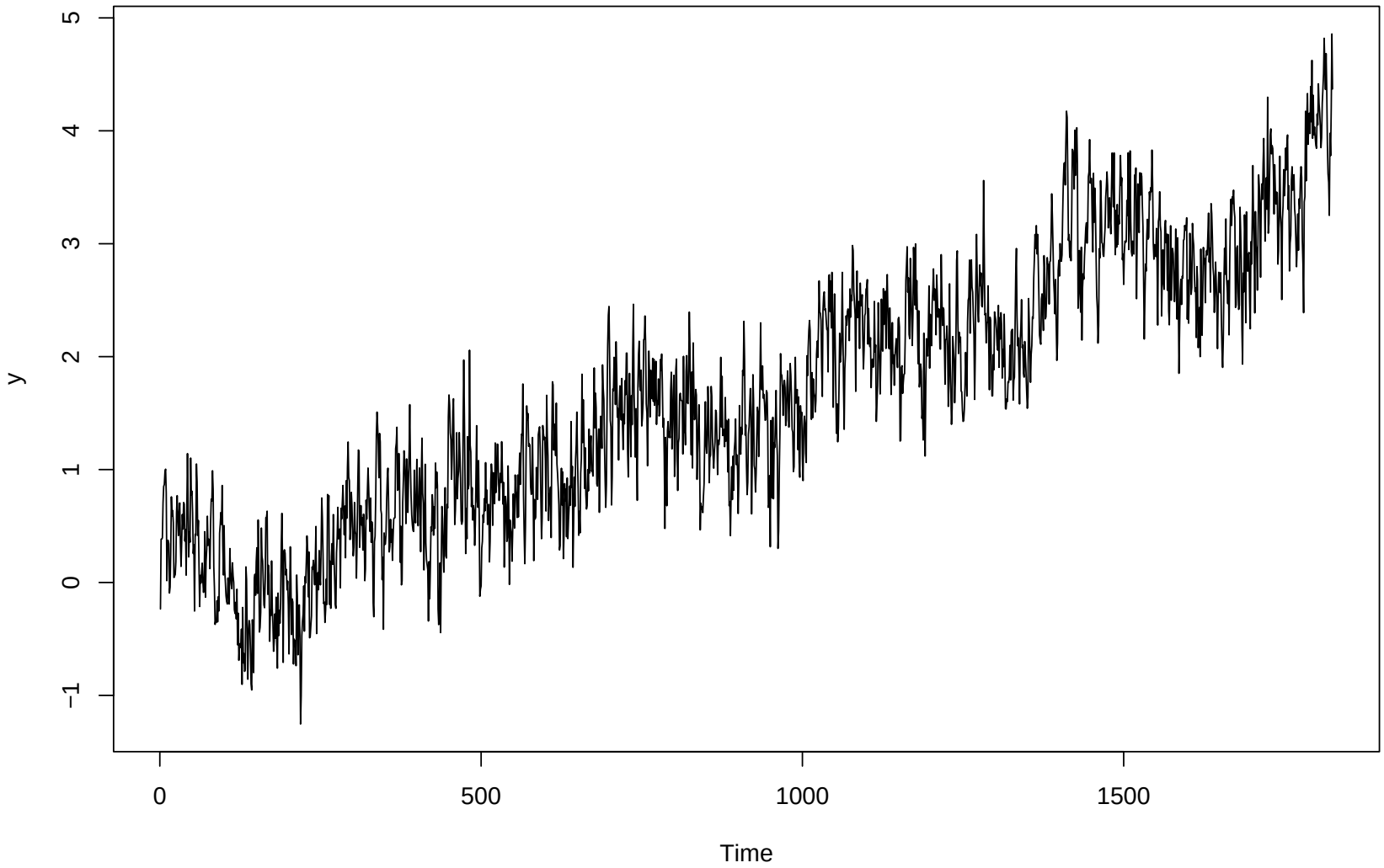


(c) $\hat{\phi} - \phi$

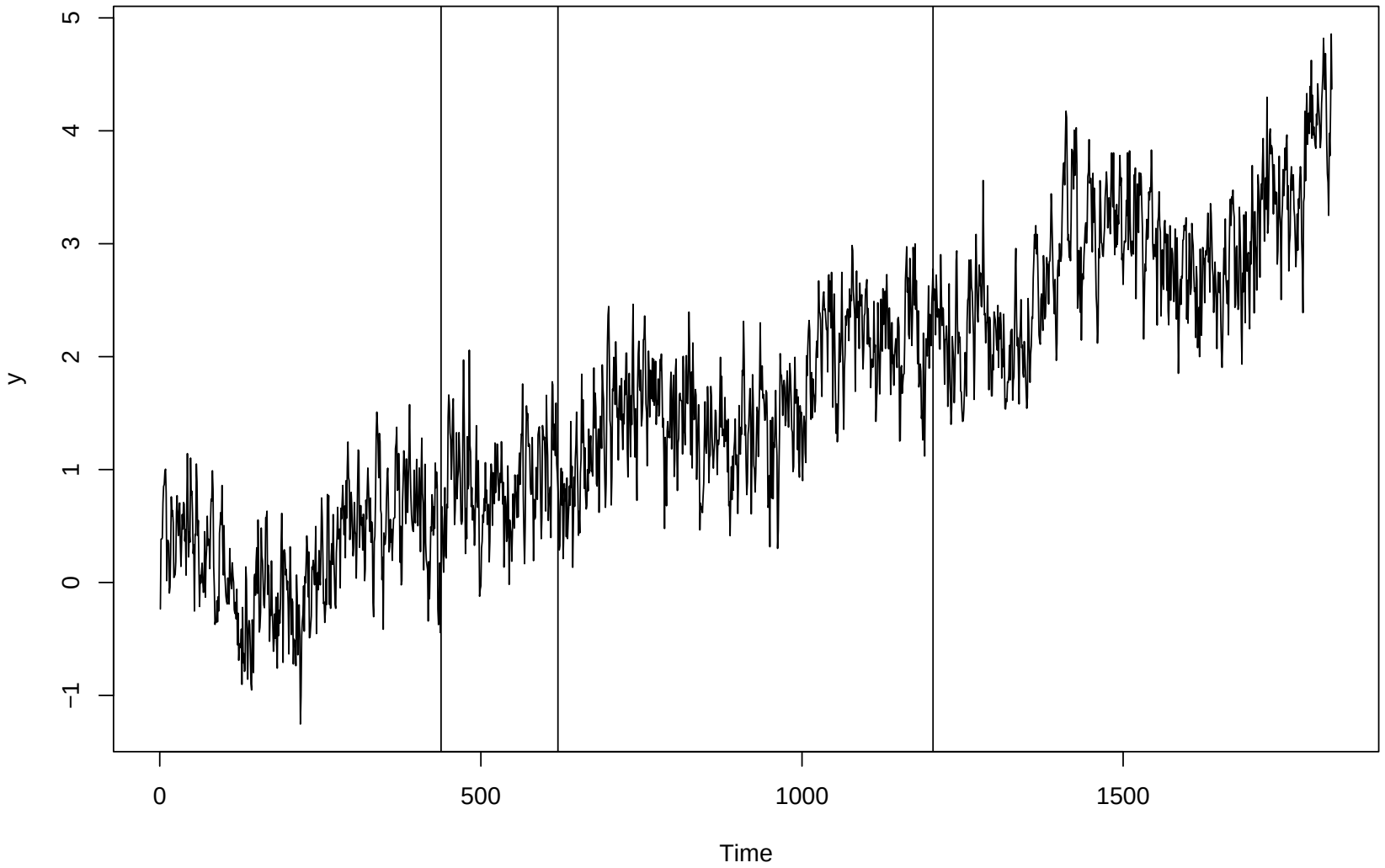


(d) $\hat{A}$

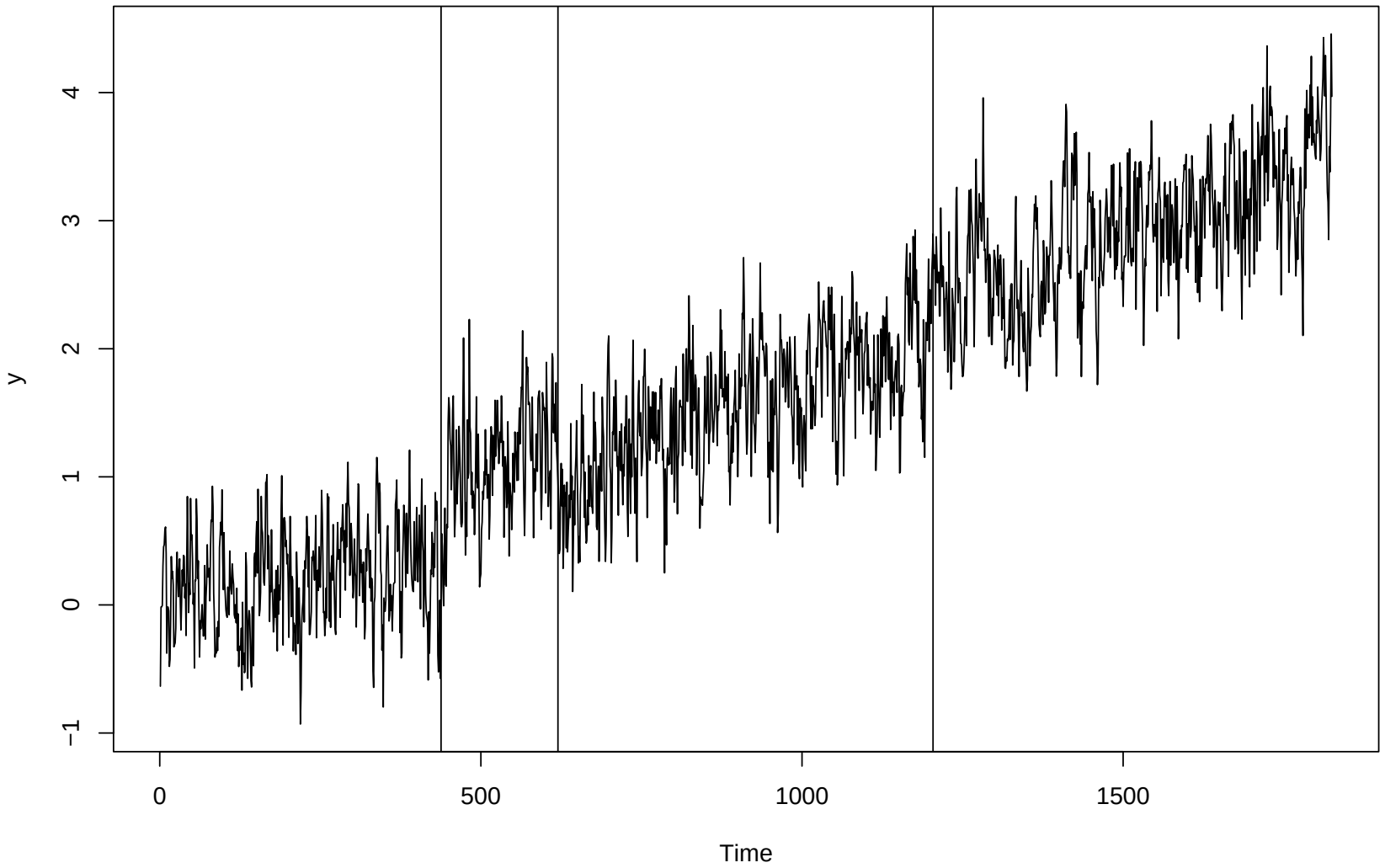
Original Example



Original Example

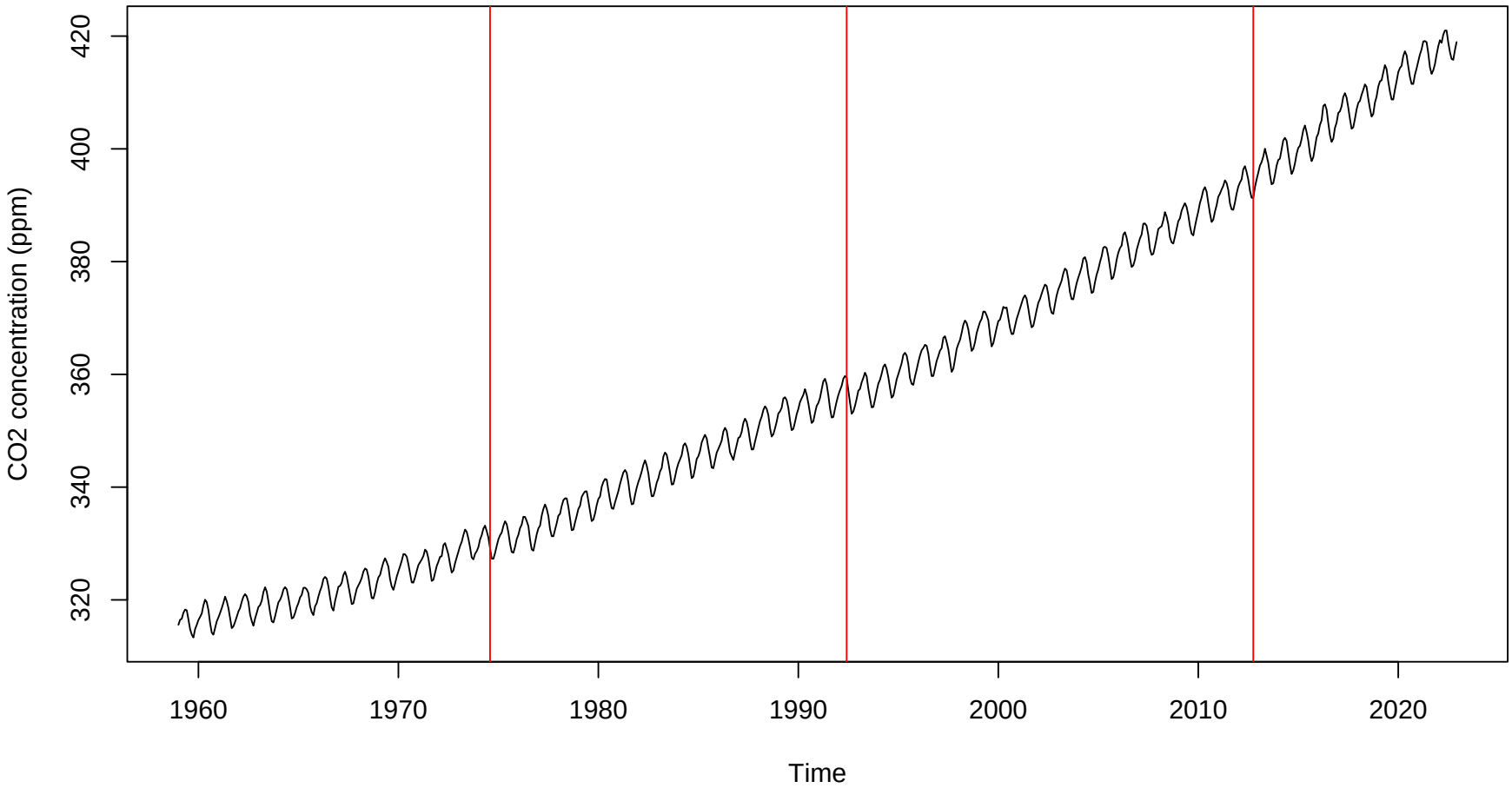


Example data



Mauna Loa

Volcanic eruption (Pinatubo) in 1991, interesting slow down in 2012.



Summary

- Changepoints occur in most data
- Detecting and documenting them improves inference
- Non-changing dynamics break additivity that search methods assume
- GD-cpt combines Gradient Descent and any existing search method to identify both local and global parameters efficiently
- This is especially important for data with a fixed seasonal cycle
- I enjoy working with messy real life data . . .
- . . . as often it sparks my next research challenge.